THE UNIVERSITY OF AUCKLAND
Te Whare Wānanga o Tāmaki Makaurau
NEW ZEALAND

ELECTENG 209

Electronics System Design

Signal Conditioning

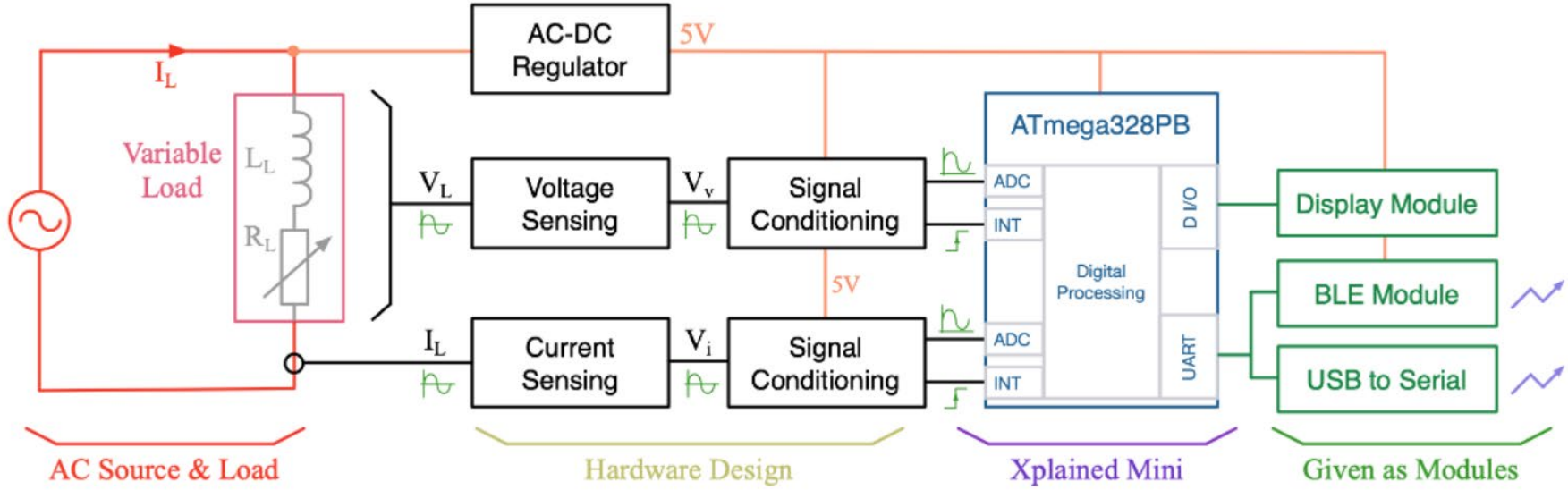
Grant Covic and Duleepa Thrimawithana

Learning Objectives

- Review of operational amplifier basics
- What are inverting and non-inverting amplifiers
- How are difference amplifiers created and why are they useful?
- Understanding practical signal conditioning circuits using single sided supplies
- What practical considerations are needed with voltage and current measurement circuits?
- How do we create RC filters and what are the design approaches in the frequency domain?
- Understanding the use and design of comparators for interrupt circuits
- Design guidelines for creating a regulated supply suited to your analogue and digital electronics.

The Energy Monitor Hardware design

- Requires Voltage Sensing, Current Sensing as discussed earlier.
- Signal conditioning
  - Level shifting, amplification and noise reduction
  - Generation of a suitable interrupt to trigger measurement from the microprocessor.
- Practical operation from a single 5V power regulator.



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Op-Amp Signal Conditioning

Review of op-amps

The inverting op-amp

The non-inverting op-amp

The difference amplifier

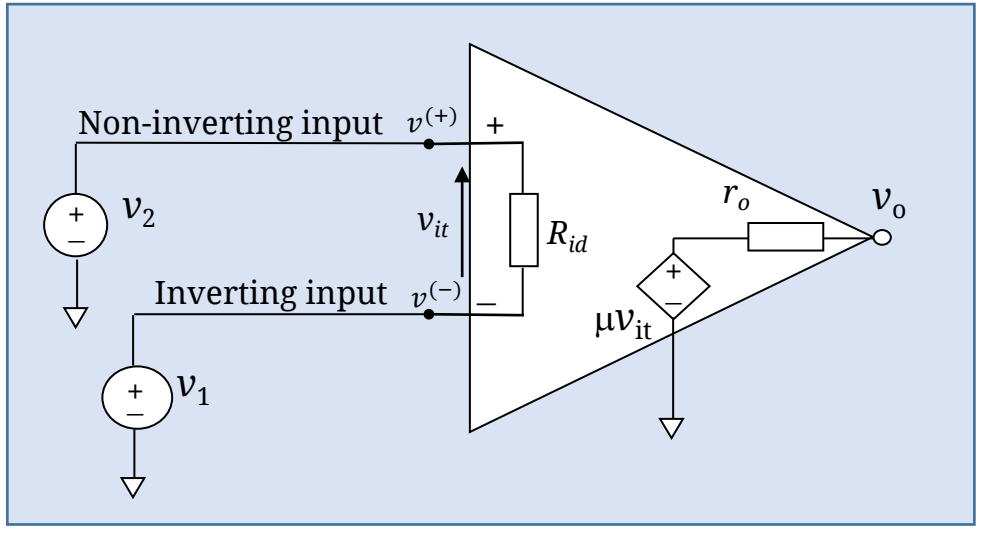
Operation from a single sided supply

Practical considerations with sensing voltage or current

The Op-Amp Equivalent Circuit

The op-amp equivalent circuit is shown in the figure below:

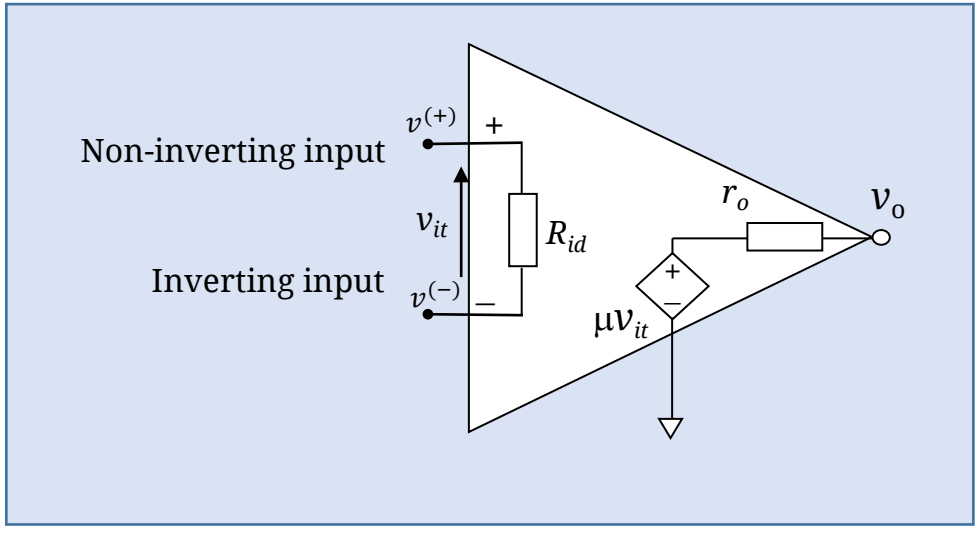
- Ideally $R_{id} = \infty,\ r_o = 0,$ and the internal gain $\mu = A = \infty$
- Input voltages v_1 and v_2 are independent
- The output voltage v_o is ideally the amplification of $v_{it} = v^{(+)} - v^{(-)}$, the voltage difference at the input terminals



Virtual Earth Principle?

- In operation the output voltage is limited by the DC supplies of the op-amp.
- If the gain of the operational amplifier is very large the input terminals are kept at the same potential. This is a ***virtual short-circuit.*** Expressing this mathematically:

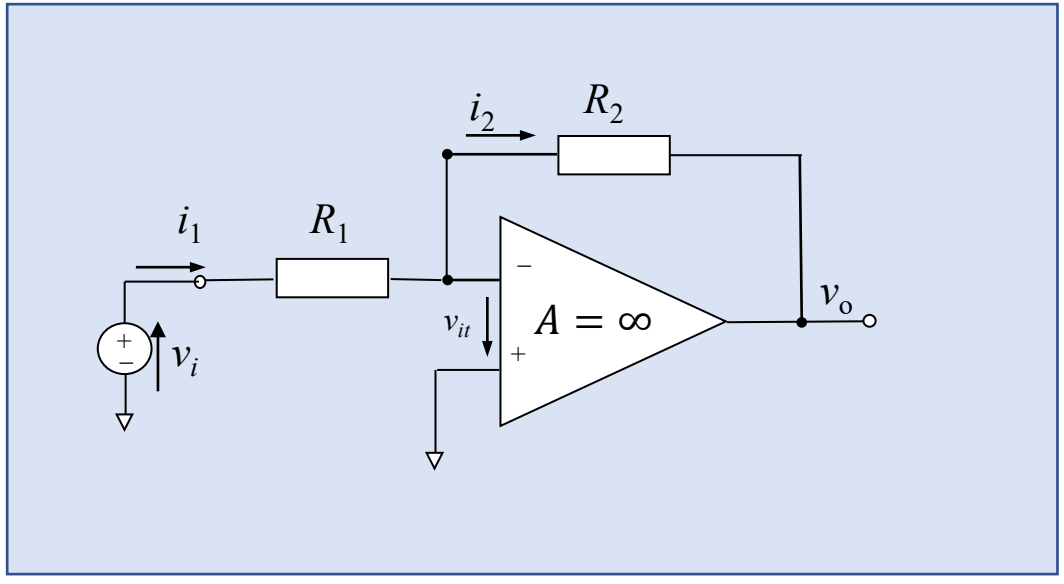
  If $A = \infty$, then the input voltage across the two terminals: $v_{it} = \left(v^{(+)} - v^{(-)}\right) = v_o/A = 0V$
- If the $v^{(+)}$terminal is "grounded" to $0V$, then this means the negative terminal $v^{(-)}$ is a ***virtual earth***.



The Inverting Operational Amplifier

A circuit can be created with a single voltage input v_i and resistors $R_1$and R_2 around this op-amp as shown below.

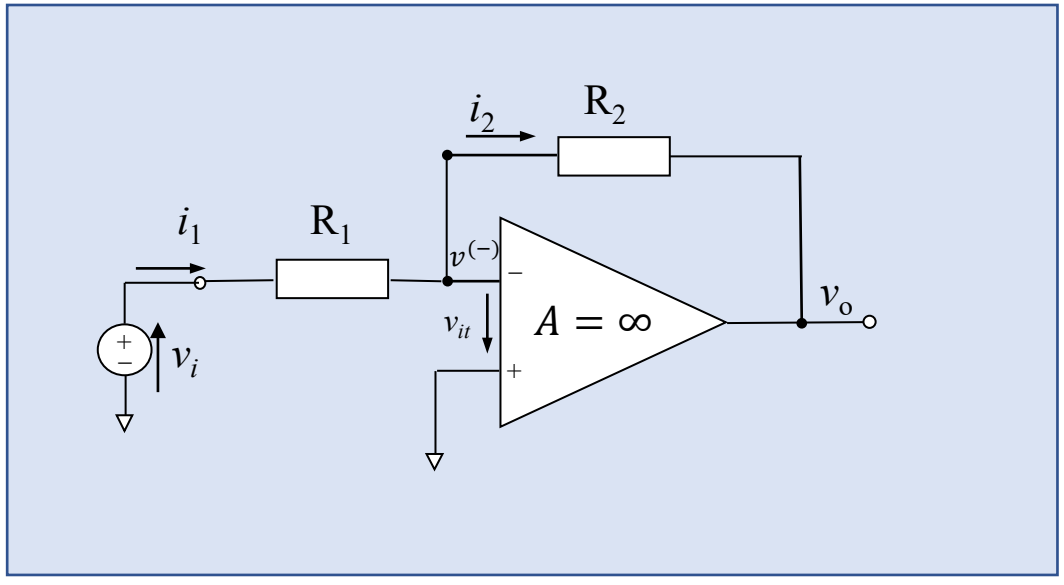
- By grounding the positive input $v^{(+)} = 0$, because of the virtual earth (short circuit at the input) effectively $v^{(-)} = 0$
- To determine the gain, use *Ohm's* and *Kirchhoff's voltage* and *current laws*.



The Inverting Operational Amplifier

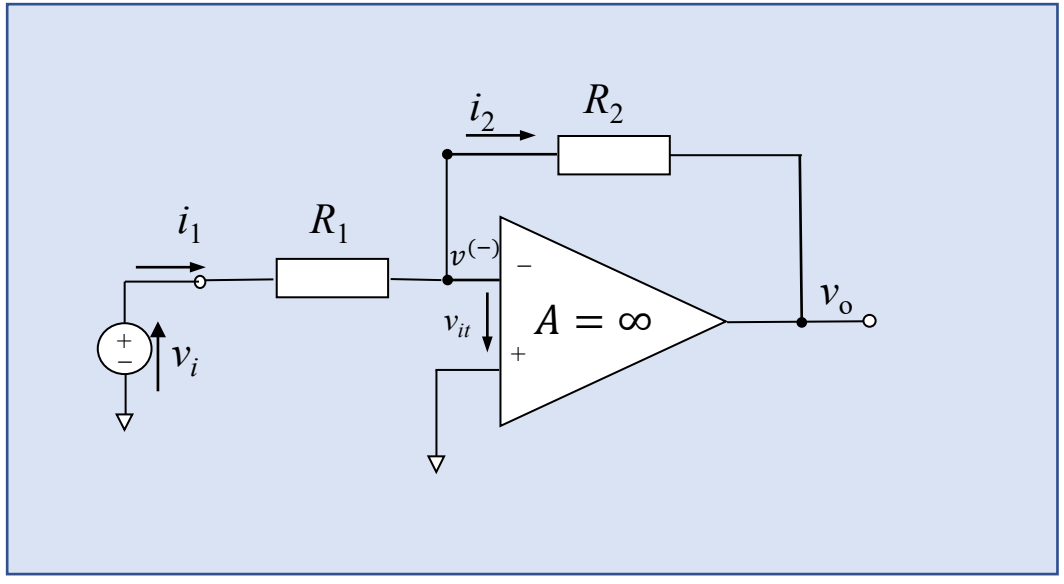
- A current i_1 will flow because of input v_i
- Using *Ohms law*, given $v^{(-)} = 0$ by virtual earth:

$$i_1 = \left(v_i - v^{(-)}\right)/R_1 = v_i/R_1$$
$$i_2 = \left(v^{(-)} - v_o\right)/R_2 = -v_o/R_2$$



The Inverting Operational Amplifier

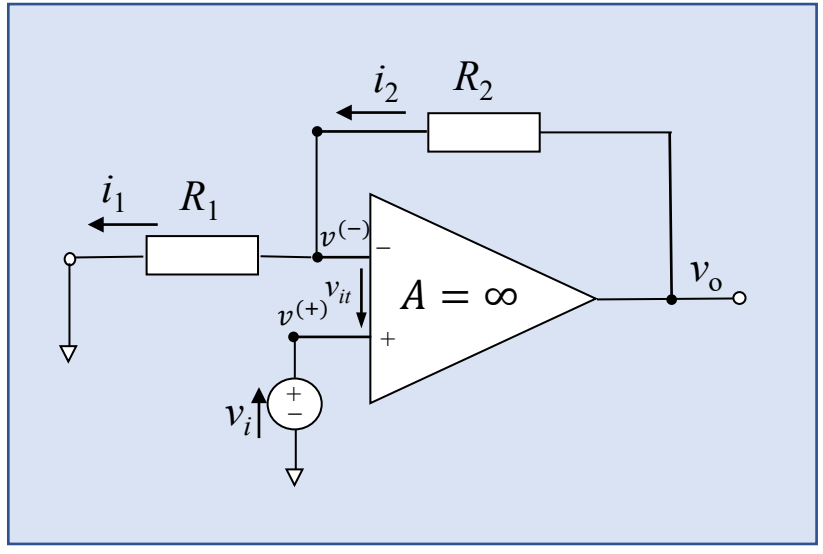
- Using *Kirchhoff's current law* at a node $\sum i = 0$, evaluate the negative input terminal node
- Since ideally $R_{id} = \infty$ then no current will flow into the op-amp so: $i_1 = i_2 \Rightarrow \frac{v_i}{R_1} = \frac{-v_o}{R_2}$
- Rearrange for the circuit voltage gain: $G = \frac{v_o}{v_i} = \frac{-R_2}{R_1}$



The Non-inverting Operational Amplifier

If the input is shifted to the positive terminal then a non-inverting configuration can be created as shown below:

- Following a similar analysis, with ideal infinite gain then $v_{it} = \left(v^{(+)} - v^{(-)}\right) = v_o/A = v_o/\infty = 0$
- Then $v^{(-)} = v^{(+)} = v_i$
- Solve for the circuit gain again using *Ohm's* and *Kirchhoff's laws*.



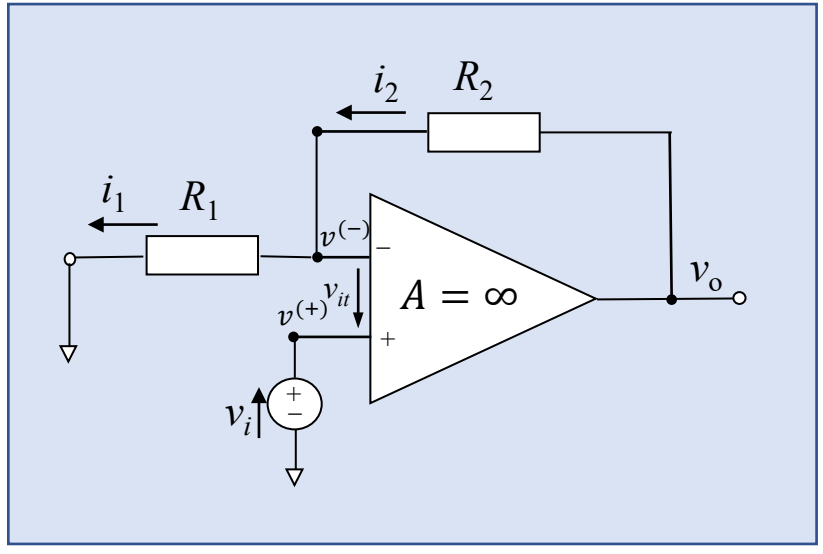
The Non-inverting Operational Amplifier

- Again the current into the op-amp at the negative terminal is zero because $R_{id} = \infty$
- So given $\sum i = 0$ then $i_2 = i_1$, where:

$$i_1 = v^{(-)}/R_1 = V_i/R_1$$

$$i_2 = \left(v_0 - v^{(-)}\right)/R_2$$

- Equating the currents and rearranging for the voltage gain gives: $G \equiv \frac{v_o}{v_i} = 1 + \frac{R_2}{R_1}$



Voltage Follower Example

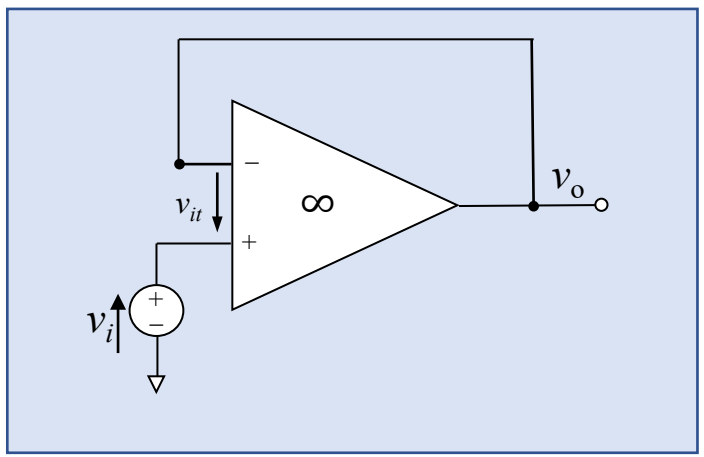
- This is simply the non-inverting configuration where:

$$R_1 = \infty\Omega$$
$$R_2 = 0\Omega$$

- The gain of the circuit is therefore

$$G \equiv \frac{v_o}{v_i} = 1 + \frac{R_2}{R_1} = 1$$

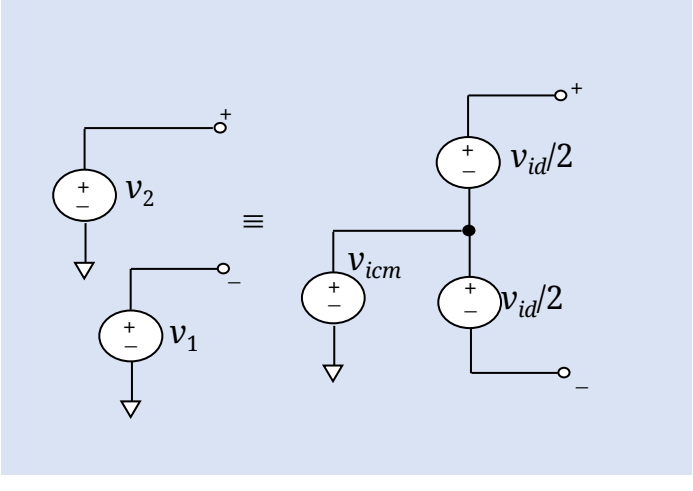
- The input impedance is the same as the op-amp, which is ideally infinite.
  This means we don't "load down" the input voltage (ideally draw zero current).



Common-mode and Difference Voltages

For circuits with two inputs, the input voltages can also be re-expressed in terms of a common (average) voltage component and a difference voltage.

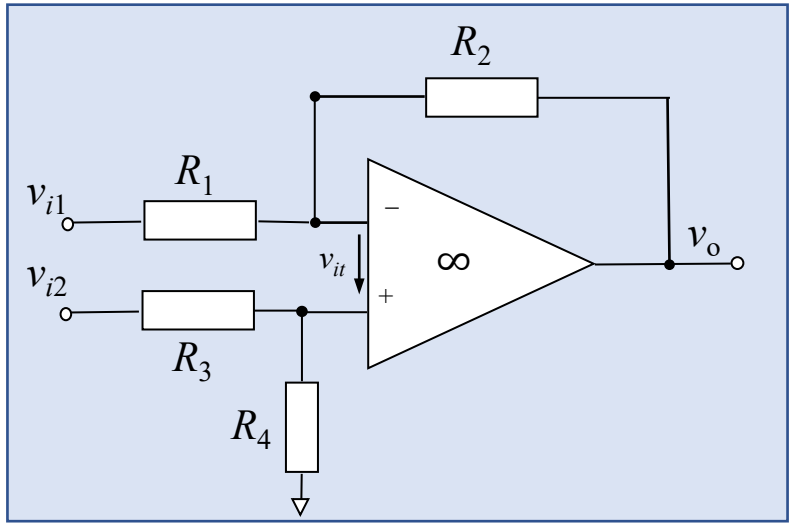
- The common (average) voltage of both inputs is called the ***common mode voltage*** : $v_{icm} = \frac{1}{2}(v_2 + v_1)$
- The ***difference voltage*** is simply : $v_{id} = v_2 - v_1$



The Single Op-Amp Difference Amplifier

Difference amplifiers (as shown below) are useful for measurement purposes in many electronic applications. While the op-amp is a difference amplifier of sorts, on its own the high gain makes it very sensitive to signal variations and non-linearities.

- Here the two input voltages are considered to be independent, and as such we can apply the principle of superposition to analyse the circuit.
- We assume $A = \infty$ (so that $v_{it} = 0$) and $R_{id} = \infty$ meaning that no current flows into the op-amp.



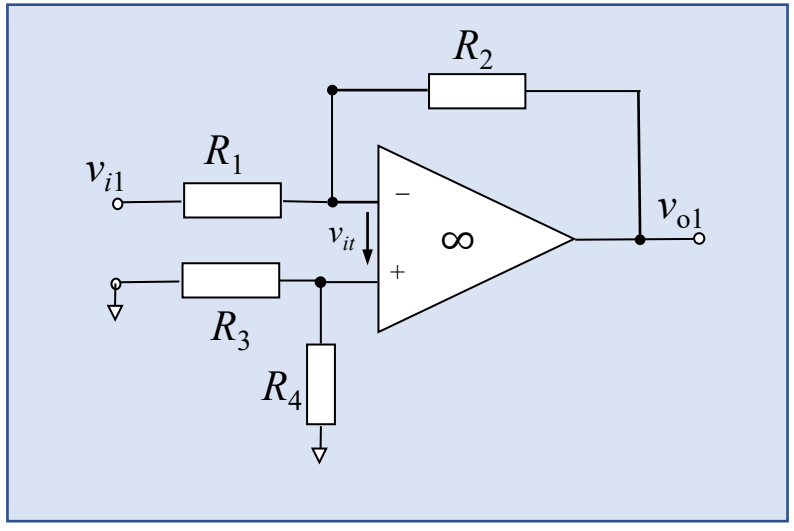
The Single Op-Amp Difference Amplifier

Reduce the second source to zero (shorting it to ground as shown) and determine the output voltage v_{o1} that results just from v_{i1}

- As shown below the voltage at the + terminal $v^{(+)} = 0$ since no current flows into the op-amp at this terminal.
- The circuit is “identical” to the inverting amplifier.

The voltage output is then:

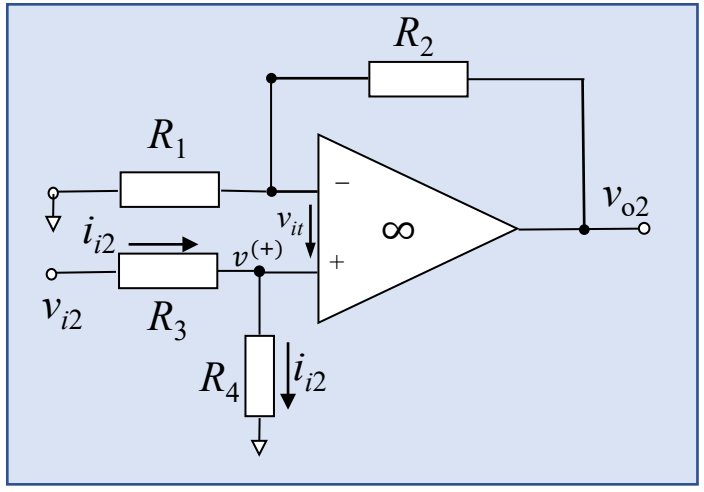
$$v_{o1} = \frac{-R_2}{R_1} v_{i1}$$



The Single Op-amp Difference Amplifier

- Reduce the first source to zero (shorting it to ground) and determine the output voltage v_{o2} that results just from v_{i2}
- The voltage at the + terminal $v^{(+)}$ can be determined by evaluating the current flow (noting no current flows into the op-amp at this terminal). This is essentially a voltage divider dependent on $R_3$and R_4.
- Thus the current i_{i2} that flows is the same in R_3 as in R_4: $i_{i2} = \frac{v_{i2}}{(R_3+R_4)} = \frac{v^{(+)}}{R_4}$

Rewriting: $v^{(+)} = \frac{R_4}{(R_3+R_4)} v_{i2} = \left(\frac{\frac{R_4}{R_3}}{1+\frac{R_4}{R_3}} \right) v_{i2}$

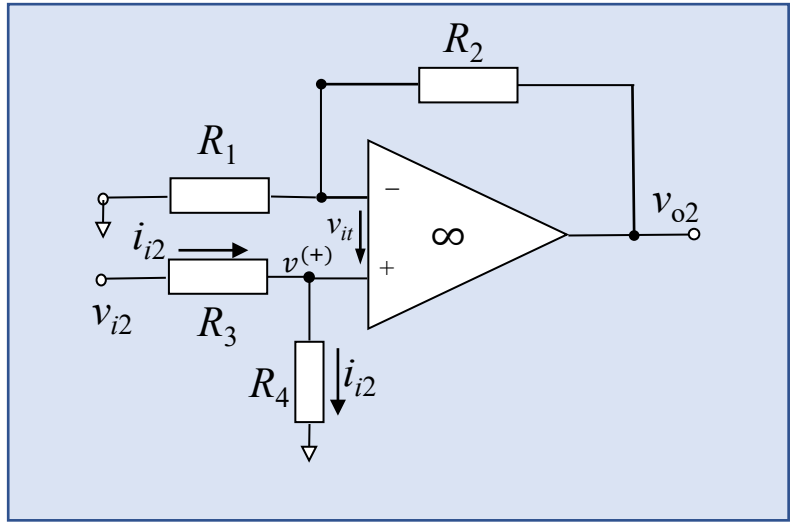


The Single Op-Amp Difference Amplifier

The circuit is identical the non-inverting amplifier, so that the voltage gain is:

$$v_{o2} = \left(1 + \frac{R_2}{R_1}\right) v^{(+)}$$

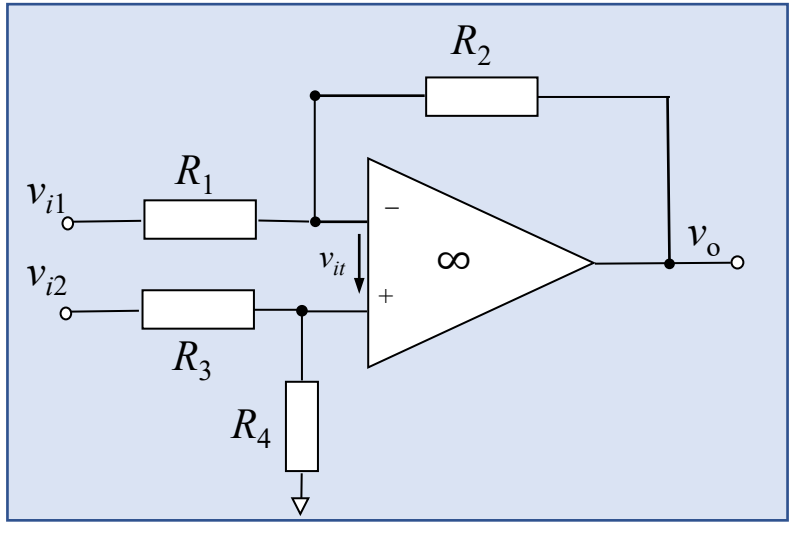
- Substituting in for $v^{(+)}$ the voltage output is : $v_{o2} = \left(1 + \frac{R_2}{R_1}\right)\left(\frac{\frac{R_4}{R_3}}{1+\frac{R_4}{R_3}}\right) v_{i2}$



The Single Op-Amp Difference Amplifier

- Finally combine both solutions: $v_o = v_{o1} + v_{o2}$, thus: $v_o = \left(1 + \frac{R_2}{R_1}\right)\left(\frac{\frac{R_4}{R_3}}{1+\frac{R_4}{R_3}}\right)v_{i2} - \frac{R_2}{R_1}v_{i1}$
- The absolute gain of each input voltage can be made identical if: $\frac{R_4}{R_3} = \frac{R_2}{R_1}$
- Thus v_o is only determined from the voltage difference of the two inputs. The gain is called the difference gain A_d:

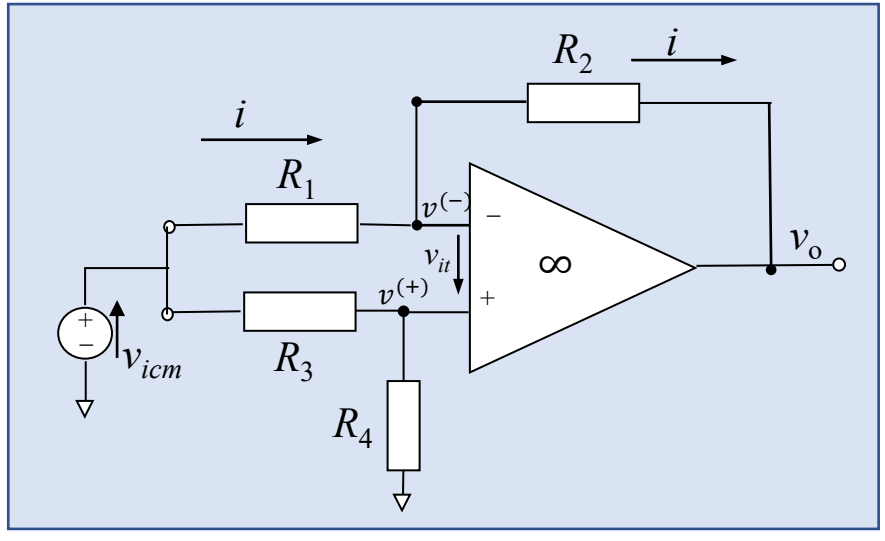
$$v_o = \left(1 + \frac{R_2}{R_1}\right)\left(\frac{\frac{R_2}{R_1}}{1 + \frac{R_2}{R_1}}\right)v_{i2} - \frac{R_2}{R_1}v_{i1} = \frac{R_2}{R_1}(v_{i2} - v_{i1}) = \frac{R_2}{R_1}v_{id} = A_d v_{id}$$



Does the Difference Amplifier have a Common Mode Gain ?

To find out, we assume both inputs are driven by the same (common) voltage, and determine the gain.

- Because the op-amp is ideal: $v^{(-)} = v^{(+)}$
- Determine $v^{(+)}$ from the input voltage as: $v^{(+)} = \frac{R_4}{(R_3+R_4)} v_{icm}$
- Now calculate current i flowing in $R_1$and R_2using *Ohms Law*



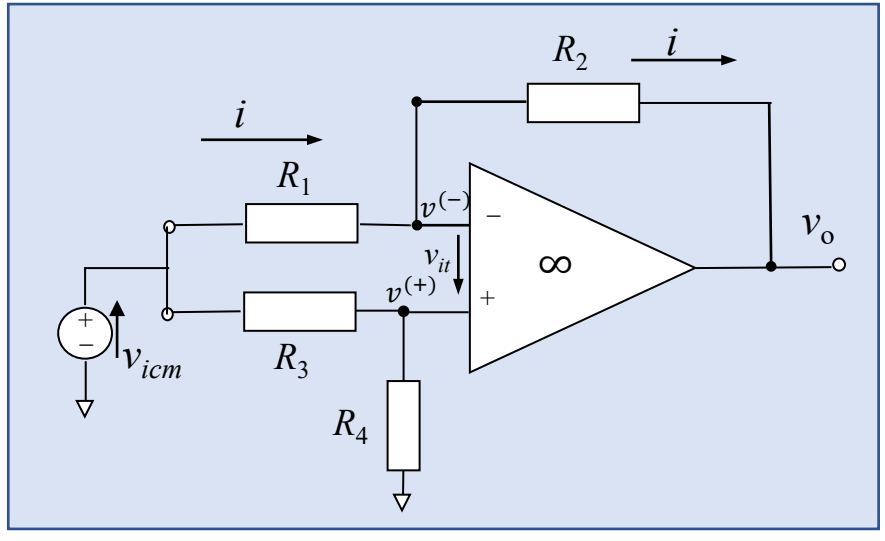
Does the Difference Amplifier have a Common Mode Gain ?

Current through R_1:

$$i = \frac{v_{icm} - v^{(-)}}{R_1} = \frac{v_{icm}\left(1 - \frac{R_4}{R_3 + R_4}\right)}{R_1} = \frac{v_{icm}}{R_1}\left(\frac{R_3}{R_3 + R_4}\right)$$

Current through R_2:

$$i = \frac{v^{(-)} - v_o}{R_2} = \frac{v_{icm}\left(\frac{R_4}{R_3 + R_4}\right) - v_o}{R_2}$$



Does the Difference Amplifier have a Common Mode Gain ?

Equate both currents:

$$\frac{1}{R_1}\left(\frac{R_3}{R_3+R_4}\right)v_{icm}=\frac{\left(\frac{R_4}{R_3+R_4}\right)v_{icm}-v_o}{R_2}\Rightarrow\frac{R_2}{R_1}\left(\frac{R_3}{R_3+R_4}\right)v_{icm}=\left(\frac{R_4}{R_3+R_4}\right)v_{icm}-v_o$$

Rearrange for v_o :

$$v_o=\left(\left(\frac{R_4}{R_3+R_4}\right)-\frac{R_2}{R_1}\frac{R_3}{R_4}\left(\frac{R_4}{R_3+R_4}\right)\right)v_{icm}=\left(\frac{R_4}{R_3+R_4}\right)\left(1-\frac{R_2}{R_1}\frac{R_3}{R_4}\right)v_{icm}=A_{cm}v_{icm}$$

If $R_4/R_3=R_2/R_1$ then $A_{cm}=0$, but if the resisters don't quite match then some common mode gain will exist.

